

ACTIVITY_MAIN.XML

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:padding="16dp"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
    <EditText
```

```
        android:id="@+id/etId"
```

```
        android:hint="Student ID"
```

```
        android:inputType="number"/>
```

```
    <EditText
```

```
        android:id="@+id/etName"
```

```
        android:hint="Student Name"/>
```

```
    <EditText
```

```
        android:id="@+id/etMarks"
```

```
        android:hint="Marks"
```

```
        android:inputType="number"/>
```

```
    <Button
```

```
        android:id="@+id/btnInsert"
```

```
        android:text="Insert"/>
```

```
    <Button
```

```
        android:id="@+id/btnUpdate"
```

```
        android:text="Update"/>
```

```
<Button  
    android:id="@+id/btnDelete"  
    android:text="Delete"/>
```

```
<Button  
    android:id="@+id/btnView"  
    android:text="View Data"/>
```

```
</LinearLayout>
```

DATABASE HELPER CLASS

```
import android.content.ContentValues
```

```
import android.content.Context
```

```
import android.database.sqlite.SQLiteDatabase
```

```
import android.database.sqlite.SQLiteOpenHelper
```

```
class DBHelper(context: Context) : SQLiteOpenHelper(context, "StudentDB", null, 1) {
```

```
    override fun onCreate(db: SQLiteDatabase) {
```

```
        db.execSQL("CREATE TABLE students(id INTEGER PRIMARY KEY, name TEXT, marks INTEGER)")
```

```
    }
```

```
    override fun onUpgrade(db: SQLiteDatabase, oldVersion: Int, newVersion: Int) {
```

```
        db.execSQL("DROP TABLE IF EXISTS students")
```

```
        onCreate(db)
```

```
    }
```

```
    fun insertData(id: Int, name: String, marks: Int): Boolean {
```

```

val db = writableDatabase
val cv = ContentValues()
cv.put("id", id)
cv.put("name", name)
cv.put("marks", marks)
val result = db.insert("students", null, cv)
return result != -1L
}

```

```

fun updateData(id: Int, name: String, marks: Int): Boolean {
    val db = writableDatabase
    val cv = ContentValues()
    cv.put("name", name)
    cv.put("marks", marks)
    val result = db.update("students", cv, "id=?", arrayOf(id.toString()))
    return result > 0
}

```

```

fun deleteData(id: Int): Boolean {
    val db = writableDatabase
    val result = db.delete("students", "id=?", arrayOf(id.toString()))
    return result > 0
}

```

```

fun getAllData(): String {
    val db = readableDatabase
    val cursor = db.rawQuery("SELECT * FROM students", null)
    var data = ""

```

```

while (cursor.moveToNext()) {
    data += "ID: ${cursor.getInt(0)}\n"
    data += "Name: ${cursor.getString(1)}\n"
    data += "Marks: ${cursor.getInt(2)}\n\n"
}
cursor.close()
return data
}
}

```

MAINACTIVITY CLASS

```

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var db: DBHelper

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        db = DBHelper(this)

        val etId = findViewById<EditText>(R.id.etId)
        val etName = findViewById<EditText>(R.id.etName)
        val etMarks = findViewById<EditText>(R.id.etMarks)
    }
}

```

```
val btnInsert = findViewById<Button>(R.id.btnInsert)
val btnUpdate = findViewById<Button>(R.id.btnUpdate)
val btnDelete = findViewById<Button>(R.id.btnDelete)
val btnView = findViewById<Button>(R.id.btnView)

btnInsert.setOnClickListener {
    val success = db.insertData(
        etId.text.toString().toInt(),
        etName.text.toString(),
        etMarks.text.toString().toInt()
    )
    Toast.makeText(this, if (success) "Inserted" else "Failed", Toast.LENGTH_SHORT).show()
}

btnUpdate.setOnClickListener {
    val success = db.updateData(
        etId.text.toString().toInt(),
        etName.text.toString(),
        etMarks.text.toString().toInt()
    )
    Toast.makeText(this, if (success) "Updated" else "Failed", Toast.LENGTH_SHORT).show()
}

btnDelete.setOnClickListener {
    val success = db.deleteData(etId.text.toString().toInt())
    Toast.makeText(this, if (success) "Deleted" else "Failed", Toast.LENGTH_SHORT).show()
}
```

```
btnView.setOnClickListener {  
    val data = db.getAllData()  
    AlertDialog.Builder(this)  
        .setTitle("Student Records")  
        .setMessage(data)  
        .setPositiveButton("OK", null)  
        .show()  
}  
}  
}
```